

M KARTHIKEYAN

OT Cybersecurity Analyst | Security Assessor | AI-Enabled Threat Detection

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🌐 **Portfolio:** karthikeyan1708.github.io — *Interactive portfolio with projects, engagements & experience*

Professional Summary

OT Cybersecurity Analyst with **1.8+ years at a Big-4 consulting firm**, delivering high-impact security assessments, firewall reviews, and compliance audits for critical-infrastructure organizations across aviation, energy, telecom, and manufacturing. Skilled in leveraging AI/ML techniques for anomaly detection and automated threat analysis. Experienced in multi-site audit execution, shift-based SOC operations, and executive-level security reporting. ISO 27001:2022 Lead Implementer certified.

Core Competencies

- **OT/ICS Security:** SCADA/DCS Assessment, Network Segmentation Review, OT Firewall Hardening, Industrial Protocol Analysis (Modbus, DNP3, OPC-UA), Physical Security Reviews
- **Security Tools:** Wireshark, Claroty, Tenable, Microsoft Sentinel, ArcStrike, Defender for IoT, Nessus, Qualys
- **Network Analysis:** Deep PCAP Analysis, Packet Inspection, Traffic Anomaly Detection, Firewall Rule Auditing, Network Architecture Review
- **Compliance & Frameworks:** ISO 27001:2022, IEC 62443, NIST CSF, CIS Controls, Maturity Scoring Models
- **Automation & Scripting:** Python (Advanced), PowerShell, Bash, Automated Report Generation
- **AI for Security:** ML-Based Anomaly Detection, NLP for Policy Analysis, Risk Scoring Engines
- **SOC Operations:** Real-Time Threat Monitoring, Alert Triage, Incident Escalation, SIEM Correlation
- **Cloud Security:** Azure Sentinel, AWS Security Hub, Cloud-Native Monitoring

Professional Experience

Associate – OT Cybersecurity

Oct 2024 – Present

PricewaterhouseCoopers (PwC), Hyderabad

1.8+ years

OT Security Assessments & Audits

- Executed **6 end-to-end OT cybersecurity assessments** across aviation, energy, telecom, and manufacturing sectors, encompassing network architecture review, control-system evaluation, and policy gap analysis.
- Led on-site security audits across **4 geographically distributed industrial sites** under aggressive 3-week timelines, delivering comprehensive maturity assessments with prioritized remediation roadmaps.
- Performed in-depth **OT firewall rule analysis and network segmentation reviews**, identifying 25+ critical misconfigurations per engagement and recommending zone-based architectures aligned with IEC 62443.
- Conducted **physical and logical security reviews** of telecom infrastructure across 5 tower locations, evaluating SMPS configurations, access controls, and environmental safeguards.
- Executed **revalidation assessments** for critical OT and IT systems, performing evidence-based control analysis and maturity scoring against ISO 27001:2022 benchmarks.

PCAP Analysis & AI-Augmented Threat Detection

- Engineered automated Python workflows to parse and analyze **multi-terabyte PCAP datasets**, extracting 50+ protocol-level features for anomaly classification.
- Developed an **ML-based anomaly detection model** processing 500K+ daily OT sensor events with 85% precision, reducing false-positive rates by 30%.
- Built an **XGBoost risk-scoring engine** that prioritized vulnerabilities by exploitability and asset criticality, cutting manual triage effort by **40%**.
- Created **NLP-based policy analysis tools** to automatically extract compliance gaps from security documentation, achieving 90%+ extraction accuracy.

SOC Operations & Continuous Monitoring

- Currently providing **24/7 shift-based SOC support** for a global industrial technology provider, performing real-time threat monitoring, alert triage, and incident escalation using Microsoft Sentinel and Defender for IoT.
- Developed correlation rules and automated playbooks to improve mean-time-to-detect (MTTD) and mean-time-to-respond (MTTR) for OT-specific threat scenarios.

Project Management & Business Development

- Managed **full project lifecycle** for a multi-phase security engagement – from kickoff through final closure, including stakeholder coordination, milestone tracking, and payment reconciliation.
- Co-authored **technical proposals** for two enterprise clients in FMCG and industrial sectors, contributing threat-landscape analysis and solution architecture sections.
- Maintained comprehensive **audit documentation**, including questionnaires, working papers, evidence trackers, and executive summary reports for partner-level review.

Technical Projects

Automatic Speech Recognition & Emotion Detection – Tamil Language *Apr–May 2024*

- Fine-tuned OpenAI Whisper with LSTM layers for emotion classification, achieving **85% accuracy** across 6 categories on a custom Tamil-speech corpus.
- Designed end-to-end data pipeline (audio augmentation, MFCC extraction, batch inference) and deployed as a containerized FastAPI service.

Communicable Disease Outbreak Prediction System *Jan–Mar 2023*

- Built RNN + SVR ensemble achieving **92% accuracy** for dengue/chikungunya forecasting over 12-month horizons using 2M+ epidemiological records.
- **Published** in Springer – *Inventive Communication and Computational Technologies* (Oct 2023).

Certifications

ISO 27001:2022 Lead Implementer – Information Security Management Systems (ISMS)

Publications

M Karthikeyan et al., “*Communicable Disease Prediction Using Machine Learning and Deep Learning Algorithms*,” *Inventive Communication and Computational Technologies*, pp. 979–992, Springer, Oct 2023.

DOI: [10.1007/978-981-99-5166-6_66](https://doi.org/10.1007/978-981-99-5166-6_66)

Education

B.Tech, Computer Science & Engineering (Artificial Intelligence) *2020–2024*

School of AI & Computing, Amrita Vishwa Vidyapeetham, Coimbatore CGPA: 8.5 / 10

Key Coursework: Network Security, Cryptography, Machine Learning, Operating Systems, Data Structures & Algorithms, Computer Networks, Database Management

Languages

English (Professional) | Tamil (Native) | Hindi (Conversational) | Kannada (Conversational)